

Kidney Transplant — Rejection (TCMR vs ABMR)



1. Rising Cr on trial

WHAT YOU SEE

A sad potato (the graft) sweating on the witness stand wired to creatinine gauges — the kidney's rising Cr is literally 'on trial', i.e. a non-specific accusation that must be proven, not a verdict of rejection.

WHAT IT ENCODES

A rising serum creatinine in a transplant patient signals graft dysfunction but is entirely NON-specific — it does not equal rejection. The differential spans pre-renal (volume/CNI vasoconstriction), intrinsic (ATN, rejection, BK nephropathy, recurrent/de novo GN, CNI toxicity), and post-renal (ureteral obstruction, lymphocele). The workflow is: confirm the rise, exclude obstruction/vascular causes with imaging, check CNI levels and BK PCR, then biopsy. Mnemonic: 'Cr is a symptom, not a diagnosis.'

BOARD TRAP

Board trap: diagnosing and treating rejection from a rising Cr alone is wrong. The Cr rise must be worked up (imaging, drug levels, BK PCR, biopsy) — empiric pulse steroids for an unexplained Cr bump can worsen CNI toxicity or BK nephropathy.

2. DDx lineup

WHAT YOU SEE

A police mugshot line-up of five suspects under the banner 'DON'T CONVICT WITHOUT EVIDENCE' — each numbered suspect is one cause of graft dysfunction, the line-up driving home that you must identify the real culprit before treating.

WHAT IT ENCODES

The core differential for acute graft dysfunction lines up as: ATN/delayed graft function (early, ischemic), CNI nephrotoxicity (high tac level, afferent vasoconstriction), BK virus nephropathy (over-immunosuppression, 1-12 months), recurrent or de novo glomerulonephritis (proteinuria, dysmorphic RBCs), and rejection (TCMR or ABMR). Surgical causes (obstruction, lymphocele, vascular thrombosis) must be excluded by ultrasound first. Each has a distinct treatment, so misclassification is harmful.

BOARD TRAP

Board trap: jumping to rejection before excluding ATN, CNI toxicity, BK nephropathy, GN, and obstruction is the central error. Several of these (CNI toxicity, BK) are treated by REDUCING immunosuppression — the opposite of rejection treatment — so 'convicting' rejection without evidence can be catastrophic.

3. CNI toxicity suspect

WHAT YOU SEE

Mugshot suspect #2, a striped tiger figure tagged 'CNI TOXICITY — afferent constriction' — the squeezing tiger stands for calcineurin-inhibitor vasoconstriction of the afferent arteriole.

WHAT IT ENCODES

Calcineurin-inhibitor nephrotoxicity comes in two forms: acute, dose-dependent, reversible afferent arteriolar vasoconstriction (Cr rises with supratherapeutic tacrolimus/cyclosporine levels) and chronic, irreversible arteriolar hyalinosis/striped interstitial fibrosis. Acutely it mimics rejection clinically but biopsy is relatively bland (isometric tubular vacuolization, no tubulitis/infiltrate) and the drug level is high. Management: reduce/adjust the CNI dose and review CYP3A4 interactions.

BOARD TRAP

Board trap: a rising Cr with a HIGH tacrolimus level and a bland biopsy is CNI nephrotoxicity → REDUCE the CNI dose. Escalating immunosuppression (pulse steroids/ATG) as if it were rejection worsens nephrotoxicity. Watch for an interacting drug (azole antifungal, diltiazem, macrolide) that raised the level.

4. BK nephropathy suspect

WHAT YOU SEE

Mugshot suspect #3, a purple virus blob tagged 'BK NEPHROPATHY — tubulitis + viral cytopathic' — the virus character marks BK polyomavirus that mimics rejection histology.

WHAT IT ENCODES

BK polyomavirus reactivates under heavy immunosuppression (peak 1-12 months) and causes interstitial nephritis/tubular injury that histologically MIMICS T-cell-mediated rejection (tubulitis, interstitial infiltrate) but shows viral cytopathic changes and positive SV40 large-T antigen immunostain; urine cytology shows 'decoy cells' and blood/urine BK PCR is high. Treatment is REDUCING immunosuppression (and screening with serial BK PCR to catch it before nephropathy).

BOARD TRAP

Board trap: treating BK nephropathy as rejection by INCREASING immunosuppression accelerates graft loss — it is the opposite of correct management (reduce IS). The discriminators are SV40+ stain, decoy cells, and high BK viral load; if tubulitis is present, prove it is rejection (not BK) before pulsing steroids.

5. Biopsy gavel

WHAT YOU SEE

A judge's wooden gavel labelled BIOPSY striking the VERDICT block — the biopsy is the gavel that delivers the only authoritative verdict (Banff diagnosis) before any sentence (treatment) is passed.

WHAT IT ENCODES

Allograft biopsy is the gold standard for diagnosing and classifying rejection using the Banff criteria; it distinguishes TCMR (tubulitis, interstitial infiltrate, +/- intimal arteritis) from ABMR (microvascular inflammation, C4d, DSA) and from non-rejection causes (ATN, CNI toxicity, BK, recurrent GN). It directs divergent therapy, so it should precede any change in immunosuppression once surgical/vascular causes are excluded by imaging.

BOARD TRAP

Board trap: treating empirically without a tissue diagnosis is the wrong answer when biopsy is feasible — the differential includes entities (CNI toxicity, BK) requiring LESS immunosuppression, so guessing can harm the graft. Image first (rule out obstruction/thrombosis), then biopsy.

6. TCMR histology

WHAT YOU SEE

Microscope slides on the upper 'TCMR' shelf showing dark crow-like lymphocytes burrowing into tubules — the invading cells are the interstitial infiltrate and tubulitis that define cellular rejection.

WHAT IT ENCODES

T-cell-mediated (acute cellular) rejection is the most common acute rejection, typically weeks to months post-transplant. Banff histology: interstitial mononuclear (lymphocytic) infiltrate, TUBULITIS (lymphocytes crossing the tubular basement membrane), and in severe (vascular) cases intimal/transmural arteritis (endarteritis). It reflects cellular alloimmunity against graft antigens and is generally reversible with prompt treatment.

BOARD TRAP

Board trap: tubulitis + interstitial infiltrate suggests TCMR, but BK nephropathy produces identical-looking tubulitis — a negative SV40 stain / low BK PCR is needed to confirm TCMR rather than BK before treating with more immunosuppression.

7. TCMR treatment

WHAT YOU SEE

A pill/steroid bottle sitting on the 'TCMR' shelf — the bottle on the cellular-rejection shelf is the pulse corticosteroid first-line treatment.

WHAT IT ENCODES

First-line treatment of acute T-cell-mediated rejection is high-dose IV pulse corticosteroids (methylprednisolone, e.g., 250-500 mg daily x3), which reverses most episodes. Steroid-resistant or severe (vascular/Banff IIB-III) TCMR is escalated to anti-thymocyte globulin (rabbit ATG) for T-cell depletion. Optimizing maintenance immunosuppression and ensuring adherence/therapeutic CNI levels are part of management.

BOARD TRAP

Board trap: plasmapheresis/IVIG/rituximab are NOT first-line for pure T-cell-mediated rejection — those treat ANTIBODY-mediated rejection. The correct sequence for TCMR is pulse steroids, then ATG if steroid-resistant. Choosing plasmapheresis for tubulitis/infiltrate (cellular) rejection is wrong.

8. ABMR histology

WHAT YOU SEE

Slides on the lower 'ABMR' shelf showing blue swollen endothelial cells stuffing the capillaries — the engorged capillaries are the microvascular inflammation/capillaritis of antibody-mediated rejection.

WHAT IT ENCODES

Antibody-mediated (humoral) rejection results from donor-specific antibodies (DSA) attacking graft endothelium, activating complement. Banff histology: microvascular inflammation — peritubular capillaritis and glomerulitis — with C4d deposition along peritubular capillaries (a footprint of complement activation), often with thrombotic microangiopathy. It can be acute or chronic (transplant glomerulopathy) and is generally more refractory than TCMR.

BOARD TRAP

Board trap: don't expect tubulitis as the hallmark of ABMR — that is the cellular (TCMR) picture. ABMR is about microvascular inflammation/capillaritis, C4d, and circulating DSA. Mislabeling capillaritis + C4d + DSA as 'cellular rejection' leads to the wrong (steroid-only) treatment.

9. C4d / DSA marker

WHAT YOU SEE

Specimen vials labelled 'C4d' and 'DSA' on the ABMR shelf — the C4d stain (complement footprint) plus circulating donor-specific antibody are the two markers that fingerprint humoral rejection.

WHAT IT ENCODES

The diagnostic triad for antibody-mediated rejection is: (1) circulating donor-specific antibodies (DSA), (2) C4d staining of peritubular capillaries (complement split product, an antibody footprint), and (3) histologic evidence of microvascular injury (glomerulitis/peritubular capillaritis). Molecular transcript assays can support the diagnosis. Pre-transplant DSA/crossmatch screening stratifies risk and guides desensitization.

BOARD TRAP

Board trap: a C4d-NEGATIVE biopsy does NOT exclude ABMR — current Banff criteria recognize C4d-negative ABMR when DSA plus microvascular inflammation (and/or molecular evidence) are present. Calling it 'not ABMR' solely because C4d is negative misses the diagnosis.

10. Hyperacute (preformed)

WHAT YOU SEE

A magnifying glass over a mottled, thrombosed kidney just released from its clamp — preformed antibodies have instantly clotted the graft microvasculature within minutes of reperfusion.

WHAT IT ENCODES

Hyperacute rejection occurs within minutes to hours of vascular clamp release, caused by PREFORMED recipient antibodies against donor ABO blood-group or HLA antigens (prior transfusions, pregnancies, transplants). Antibody binding triggers complement, endothelial injury, and diffuse microvascular thrombosis → the graft mottles and is lost; explantation is required. It is essentially untreatable once it starts.

BOARD TRAP

Board trap: hyperacute rejection is PREVENTED, not treated — by ABO matching and a pre-transplant lymphocytotoxic crossmatch (and now virtual crossmatch/DSA screening). A stem offering 'pulse steroids' or 'plasmapheresis' to salvage an intra-operative hyperacute reaction is wrong; the graft is lost and the lesson is prevention via crossmatch.

11. Chronic rejection

WHAT YOU SEE

A small, shrunken, scarred kidney drawn off to the side — the atrophic fibrotic graft is the slow, irreversible interstitial fibrosis/tubular atrophy of chronic rejection.

WHAT IT ENCODES

Chronic allograft injury develops over months to years as an indolent rise in Cr with proteinuria and worsening hypertension. Pathology shows interstitial fibrosis and tubular atrophy (IF/TA), transplant glomerulopathy (duplicated GBM), and chronic vasculopathy (intimal fibrosis). It is driven by chronic antibody-mediated injury (chronic ABMR/DSA), recurrent subclinical rejection, CNI toxicity, and nonimmune factors (HTN, recurrent disease). Damage is largely irreversible.

BOARD TRAP

Board trap: chronic rejection/IF-TA is NOT reversed by pulse steroids or ATG — choosing acute-rejection therapy for an indolent months-to-years fibrotic decline is wrong. Management is supportive/preventive: optimize BP, minimize CNI toxicity, control DSA, ensure adherence; acute rescue regimens do not undo established fibrosis.

12. ABMR treatment

WHAT YOU SEE

Boards-band icons reading plasmapheresis + IVIG + rituximab + bortezomib — the four-drug toolkit that removes and blocks the antibody/B-cell axis driving ABMR.

WHAT IT ENCODES

Treatment of antibody-mediated rejection targets the antibody/B-cell axis: plasmapheresis (remove circulating DSA) + IVIG (immunomodulate/neutralize), often with rituximab (anti-CD20 B-cell depletion) and sometimes bortezomib (proteasome inhibitor against plasma cells) or eculizumab (terminal complement blockade) in refractory cases. Pre-transplant desensitization for high-DSA recipients uses the same principles. ABMR is generally harder to reverse than TCMR.

BOARD TRAP

Board trap: steroids ALONE are inadequate for ABMR — the antibody must be removed/blocked (plasmapheresis + IVIG ± rituximab/bortezomib). Treating a C4d+/DSA+ humoral rejection with pulse methylprednisolone only (the TCMR recipe) is the classic wrong answer.

13. Steroid-resistant → ATG

WHAT YOU SEE

A rabbit icon (rabbit ATG) placed after a crossed-out steroid bottle — once pulse steroids fail, the rabbit-derived anti-thymocyte globulin is the next step for cellular rejection.

WHAT IT ENCODES

When acute T-cell-mediated rejection fails to respond to high-dose pulse corticosteroids (steroid-resistant TCMR) or is severe/vascular at presentation, the next step is rabbit anti-thymocyte globulin (ATG), which profoundly depletes T cells. ATG necessitates intensified infection prophylaxis (CMV with valganciclovir, PJP with TMP-SMX) because of the resulting deep immunosuppression and reactivation risk.

BOARD TRAP

Board trap: escalate steroid-resistant CELLULAR rejection to ATG — not to plasmapheresis/rituximab (those are for ANTIBODY-mediated rejection). Also remember ATG triggers CMV reactivation, so a febrile, leukopenic patient after ATG needs a CMV PCR, not automatic re-treatment for 'persistent rejection.'

14. BK → reduce IS

WHAT YOU SEE

A rising-Cr arrow plus a 'BK' label both pointing to 'REDUCE IMMUNOSUPPRESSION' — BK does the OPPOSITE of rejection: dial the immunosuppression down, not up.

WHAT IT ENCODES

BK polyomavirus nephropathy is managed by REDUCING immunosuppression (typically lower the antimetabolite/CNI), allowing the patient's immune system to clear the virus, guided by serial BK PCR. There is no reliably effective antiviral; cidofovir/leflunomide/IVIG are adjuncts of limited evidence. Routine screening with blood BK PCR allows pre-emptive IS reduction before nephropathy develops.

BOARD TRAP

Board trap: BK nephropathy is the inverse of rejection — give LESS immunosuppression, not more. Because BK histology mimics TCMR (tubulitis), the dangerous error is pulsing steroids/intensifying IS, which fuels viral replication and graft loss. Confirm with SV40 stain/decoy cells/high BK PCR and reduce IS.